Mathematical Foundations of A/B Testing

Probability Models, Hypothesis Testing, and Confidence Intervals

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1 Derivations

1.1 Definition and Notation

A/B testing compares a control group with a variant group to determine whether a modification changes the probability of a desired outcome, such as a user making a purchase.

Let p_1 and p_2 denote the population conversion probabilities of the control and variant groups, respectively. Our parameter of interest is

$$\Delta = p_1 - p_2.$$

Suppose the groups contain n_1 and n_2 independent users, respectively. Each user has a binary outcome (conversion or no conversion), with a constant conversion probability within each group. We assume independence within and between groups.

1.2 Probability Model and Maximum Likelihood Estimation

For user i in the control group and user j in the variant group, define the indicator random variables

$$X_{1i} = \begin{cases} 1, & \text{if user } i \text{ converts,} \\ 0, & \text{otherwise,} \end{cases} \qquad X_{2j} = \begin{cases} 1, & \text{if user } j \text{ converts,} \\ 0, & \text{otherwise.} \end{cases}$$

Assuming independent outcomes with constant conversion probabilities within each group,

$$X_{1i} \sim \text{Bernoulli}(p_1), \qquad X_{2j} \sim \text{Bernoulli}(p_2).$$

Let $S_1 = \sum_{i=1}^{n_1} X_{1i}$ and $S_2 = \sum_{j=1}^{n_2} X_{2j}$ denote the numbers of conversions. Then

$$S_1 \sim \text{Binomial}(n_1, p_1), \qquad S_2 \sim \text{Binomial}(n_2, p_2).$$

Writing s_1, s_2 for the observed conversion counts, independence gives the likelihood

$$L((p_1, p_2); S_1 = s_1, S_2 = s_2) = \binom{n_1}{s_1}\binom{n_2}{s_2} \cdot p_1^{s_1}(1-p_1)^{n_1-s_1} \cdot p_2^{s_2}(1-p_2)^{n_2-s_2}.$$

The log-likelihood separates into terms involving p_1 and p_2. Differentiating partially with respect to p_1 and setting the derivative to zero gives

$$\frac{\partial \log L}{\partial p_1} = \frac{s_1}{p_1} - \frac{n_1 - s_1}{1 - p_1} = 0,$$

which yields $\widehat{p}_1 = s_1/n_1$. Similarly,

$$\frac{\partial \log L}{\partial p_2} = \frac{s_2}{p_2} - \frac{n_2 - s_2}{1 - p_2} = 0 \quad \Longrightarrow \quad \widehat{p}_2 = \frac{s_2}{n_2}.$$

Both estimates maximize the likelihood.(MLE).

1.3 Two-Proportion Z-Test

We test

$$H_0 : p_1 = p_2 \qquad \text{against} \qquad H_1 : p_1 \neq p_2.$$

Let $D = \widehat{p}_1 - \widehat{p}_2$. Since

$$E[\widehat{p}_k] = E[\frac{S_k}{n_k}] = \frac{n_k p_k}{n_k} = p_k$$

and

$$\text{Var}(\widehat{p}_k) = \frac{\text{Var}(S_k)}{n_k^2} = p_k(1-p_k)/n_k$$

, independence gives

$$E[D] = p_1 - p_2, \qquad \text{Var}(D) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}.$$

Under the null hypothesis $H_0 : p_1 = p_2$, the expected difference $E[D]$ is zero. However, the observed difference may not be exactly zero because of random sampling variation. We therefore determine whether the observed difference is unusually large relative to the variation expected under H_0.

Under H_0, let $p_1 = p_2 = p$. Since p is unknown, we estimate it using the pooled

estimator

$$\widehat{p} = \frac{s_1 + s_2}{n_1 + n_2},$$

where s_1 and s_2 are the observed conversion counts. Substituting this estimate into the variance under H_0 gives the pooled standard error:

$$\widehat{\mathrm{SE}}_0 = \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}.$$

We define the test statistic

$$\boxed{Z = \frac{\widehat{p}_1 - \widehat{p}_2}{\widehat{\mathrm{SE}}_0}.}$$

Thus, Z measures how many standard errors the observed difference lies from zero. By the Central Limit Theorem, for sufficiently large sample sizes, Z is approximately standard normal under H_0.

The p-value measures how unusual the observed result is under the assumption that H_0 is true. Since the alternative hypothesis is two-sided, we consider extreme values in both directions:

$$\boxed{p\text{-value} \approx 2\left[1 - \Phi(|z_{\mathrm{obs}}|)\right],}$$

where z_{obs} is the observed value of Z and Φ is the standard normal cumulative distribution function.

We choose a significance level α before conducting the test. It represents the tolerated probability of a Type I error, namely rejecting H_0 when it is true. The critical value $z_{1-\alpha/2}$ satisfies

$$P\left(|Z| > z_{1-\alpha/2} \mid H_0\right) \approx \alpha.$$

Hence, the rejection rule is

$$\boxed{\text{Reject } H_0 \text{ if } |z_{\mathrm{obs}}| > z_{1-\alpha/2}.}$$

Equivalently, we reject H_0 if the p-value is less than α.